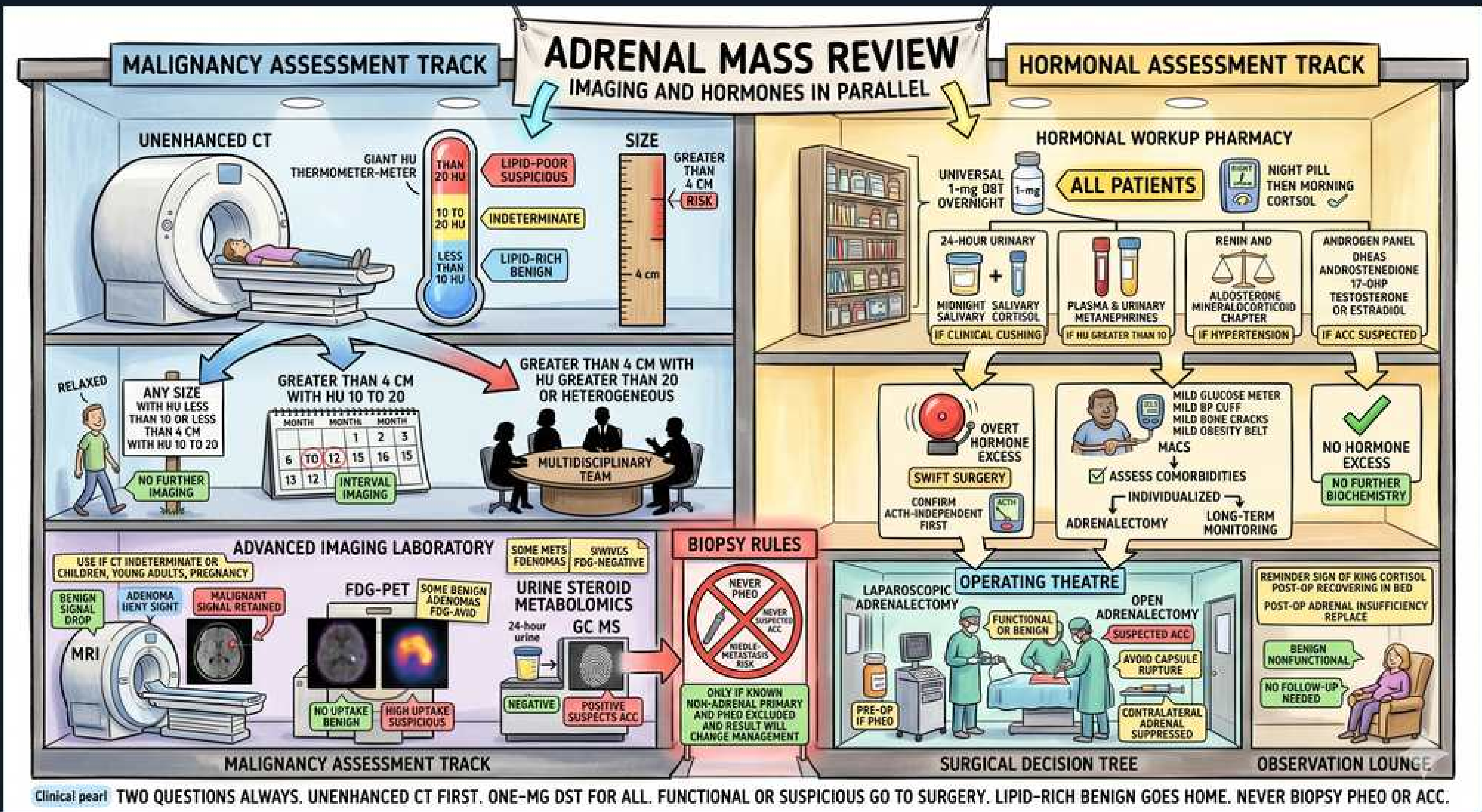
Adrenal — Mass Review: Imaging & Surgical Decision Tree



1. Two questions always

WHAT YOU SEE

Title banner 'ADRENAL MASS REVIEW — IMAGING AND HORMONES IN PARALLEL' spanning two side-by-side tracks (a CT/imaging wing and a hormone pharmacy wing) — the split layout literally shows the two questions being asked in parallel: is it malignant, and is it functional.

WHAT IT ENCODES

For every adrenal incidentaloma ask two questions in PARALLEL: (1) is it FUNCTIONAL (autonomous hormone secretion)? and (2) is it MALIGNANT (imaging phenotype + size)? Imaging and biochemistry are done together, and either a functional or a suspicious result pushes toward surgery.

2. Unenhanced CT first

WHAT YOU SEE

CT scanner, HU thermometer '≤10 lipid-rich benign, >20 suspicious'

WHAT IT ENCODES

Non-contrast CT attenuation is the first-line malignancy discriminator: ≤10 HU = lipid-rich benign adenoma (no further imaging); 10–20 HU indeterminate (use washout or MRI chemical-shift); >20 HU or heterogeneous = suspicious. On a washout study an adenoma shows absolute washout >60% or relative washout >40%.

BOARD TRAP

≤10 HU is the benign value; the reversal trap is reading high HU as reassuring — >20 HU is the worrying one.

3. Size >4cm

WHAT YOU SEE

Size ruler 'GREATER THAN 4 CM'

WHAT IT ENCODES

Size raises malignancy probability: <4 cm usually benign; >4 cm is a surgical threshold trigger; >6 cm carries high ACC risk and is generally resected outright. Interval growth >0.8–1 cm on follow-up is also a red flag.

4. Lipid-rich benign → home

WHAT YOU SEE

'ANY SIZE WITH HU ≤10 — NO FURTHER WORKUP'

WHAT IT ENCODES

A lipid-rich adenoma (non-contrast ≤10 HU), if non-functional, needs NO further imaging characterization — the low attenuation is specific enough for benignity. (It still gets the one-time hormonal screen.)

BOARD TRAP

Don't over-image or biopsy a clearly lipid-rich (≤10 HU) lesion — additional imaging adds nothing once HU confirms benign.

5. >4cm + HU>20/heterogeneous → MDT

WHAT YOU SEE

'GREATER THAN 4CM WITH HU>20 OR HETEROGENEOUS — MULTIDISCIPLINARY TEAM'

WHAT IT ENCODES

>4 cm with HU >20, heterogeneity, irregular margins, or invasion = suspicious for ACC → refer to a multidisciplinary team and lean toward RESECTION (open adrenalectomy if ACC suspected). FDG-PET/MRI may further characterize before surgery.

6. Hormonal workup — all patients

WHAT YOU SEE

'HORMONAL ASSESSMENT TRACK — ALL PATIENTS' pharmacy

WHAT IT ENCODES

Every adrenal mass — regardless of size — gets a biochemical screen for autonomous hormone excess: cortisol (1 mg overnight DST), pheochromocytoma (metanephrines), and, if hypertensive/hypokalemic, aldosterone:renin ratio. Add adrenal androgens (DHEAS) when virilization/ACC is suspected.

BOARD TRAP

Hormonal screening is universal — skipping the workup on a 'small benign-looking' mass misses subclinical/autonomous cortisol secretion and pheo.

7. 1mg DST overnight

WHAT YOU SEE

'UNIVERSAL 1MG DST OVERNIGHT' for all

WHAT IT ENCODES

The 1 mg overnight dexamethasone suppression test is universal: give 1 mg dexamethasone at 11 pm–midnight, measure 8 am cortisol. Failure to suppress to ≤1.8 μg/dL (50 nmol/L) indicates possible autonomous cortisol secretion (MACS/subclinical Cushing); >5 μg/dL is more definite. Confirm with ACTH (low/suppressed) and late-night salivary cortisol / 24-h urinary free cortisol.

BOARD TRAP

Suppression to ≤1.8 μg/dL excludes autonomous cortisol; a value above this is NOT normal — failure to suppress is the abnormal/positive screen, not the reassuring one.

8. Plasma/urine metanephrines (pheo)

WHAT YOU SEE

'PLASMA & URINARY METANEPHRINES' pheo screen

WHAT IT ENCODES

Screen for pheochromocytoma with plasma free metanephrines or 24-h urinary fractionated metanephrines (high sensitivity). This is mandatory before any biopsy or surgery; a positive screen mandates preoperative alpha-blockade (phenoxybenzamine/doxazosin) THEN beta-blockade, with volume/salt loading.

BOARD TRAP

Never give a beta-blocker first in pheo (unopposed alpha → hypertensive crisis) — alpha-block before beta-block.

9. Renin/aldosterone if HTN

WHAT YOU SEE

'RENIN & ALDOSTERONE RATIO IF HTN' / hypokalemia

WHAT IT ENCODES

In patients with hypertension and/or hypokalemia, screen for primary aldosteronism with the aldosterone:renin ratio (ARR) — high aldosterone with suppressed renin. Confirm with a saline/captopril suppression test, then lateralize with adrenal vein sampling before unilateral adrenalectomy.

BOARD TRAP

Stopping interfering drugs matters — MRAs (spironolactone/eplerenone) and other agents alter the ARR; also a unilateral mass on CT may not be the aldosterone source, so AVS lateralization is needed before surgery.

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10. Androgen / DHEAS panel

WHAT YOU SEE

'ANDROGEN PANEL — DHEAS, 17-OH...' 17-testosterone bottle

WHAT IT ENCODES

Measure adrenal androgens (DHEAS, testosterone, 17-OH-progesterone) when there is virilization, rapid hirsutism, or a large/suspicious mass. Markedly elevated DHEAS/androgens with a large heterogeneous mass strongly suggest adrenocortical carcinoma (or, with high 17-OHP, congenital adrenal hyperplasia).

BOARD TRAP

New virilization with high DHEAS + a large adrenal mass = ACC until proven otherwise — don't write it off as a benign androgen-secreting adenoma or PCOS.

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11. Functional or suspicious → surgery

WHAT YOU SEE

'OVERT HORMONE EXCESS — SWIFT SURGERY', adrenalectomy

WHAT IT ENCODES

Surgical indications for an incidentaloma: overt hormone excess (functional — Cushing, pheo, aldosteronoma, virilizing ACC), size >4 cm, or imaging-suspicious for malignancy. Adrenalectomy is then the treatment; correct the hormonal milieu and pretreat pheo before the OR.

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12. FDG-PET / advanced imaging

WHAT YOU SEE

'ADVANCED IMAGING LABORATORY — FDG-PET, MRI' for indeterminate

WHAT IT ENCODES

For indeterminate lesions (10–20 HU, equivocal washout), use MRI chemical-shift (signal drop-out on out-of-phase = lipid = adenoma) or FDG-PET (avid = malignant/ACC/metastasis). PET also helps stage ACC and find metastases.

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13. Biopsy only after excluding pheo & known cancer

WHAT YOU SEE

Red 'BIOPSY RULES — never biopsy unless pheo excluded'

WHAT IT ENCODES

Biopsy is almost never indicated: it cannot diagnose ACC (needs whole-specimen Weiss score) and risks needle-track seeding/capsule violation, and it can trigger crisis in pheo. The ONLY legitimate scenario is a suspected metastasis in a patient with a known extra-adrenal cancer — and only after pheo is biochemically excluded.

BOARD TRAP

Never biopsy before excluding pheochromocytoma; never biopsy a suspected ACC at all.

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14. Lap vs open adrenalectomy

WHAT YOU SEE

'OPERATING THEATRE — laparoscopic vs OPEN for suspected ACC'

WHAT IT ENCODES

Laparoscopic adrenalectomy is preferred for benign/functioning lesions and small masses. OPEN adrenalectomy is mandated for suspected ACC or any mass with local/vascular invasion — to achieve en-bloc R0 resection and avoid capsule rupture/tumor spillage that worsens prognosis.

BOARD TRAP

Don't take a suspected ACC laparoscopically — capsule violation/spillage seeds tumor and ruins the only curative chance; go open.

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15. Observation lounge

WHAT YOU SEE

'OBSERVATION LOUNGE — no follow-up if benign nonfunctional'

WHAT IT ENCODES

A benign-appearing (≤10 HU, <4 cm, homogeneous), non-functional adenoma can be reassured with minimal or no routine follow-up per current guidelines. If any feature is indeterminate or borderline-functional, individualized repeat imaging/hormone testing is reasonable.

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16. Post-op adrenal insufficiency risk

WHAT YOU SEE

Right-bottom sign 'POST-OP: WATCH FOR ADRENAL INSUFFICIENCY'

WHAT IT ENCODES

After unilateral adrenalectomy for a cortisol-secreting (or autonomously secreting) adenoma, the chronically suppressed contralateral gland/HPA axis may not immediately resume output → transient secondary adrenal insufficiency. Cover with perioperative glucocorticoids and taper slowly with morning cortisol/ACTH-stimulation guidance over months.

BOARD TRAP

Don't forget post-resection adrenal insufficiency in a patient who had autonomous cortisol secretion — stopping steroids abruptly can precipitate adrenal crisis.
